

Learning from HS2 and applying within Network Rail

Promoting Engineering Excellence

Niall Fagan, Head of Engineering
Capital Delivery Eastern



The need for Engineering Excellence..



- Cost of living crises
- £60bn black hole in public finances
- Political uncertainty
- War in Ukraine and energy crises
- And what has it got to do with Engineering Excellence

Everything!!



Moody's
UK economic outlook
downgraded to
'negative' by rating
agency

**Moody's say downgrade from 'stable'
was driven by political instability and
high inflation**

● **Tory leadership contest - latest
updates**

PA Media

Sat 22 Oct 2022 08.28 BST



The UK's economic outlook has been
downgraded from "stable" to
"negative" by the rating agency



So, what can Engineers do...?

- Question, understand, reason, challenge, produce... Take ownership
- Utilise engineering principles to challenge current practices and standards
- Can you make a career out of **following process**...?



What I will cover this evening

- HS2 approach to specification and design
- Potential failure mechanisms of track
- How we can challenge and improve (learning from both good practice and failure)
- Technical Issues
 - Track system design: slab v ballasted track
 - Track alignment
 - S&C
- Consider Line speed Improvements on 140mph on ECML



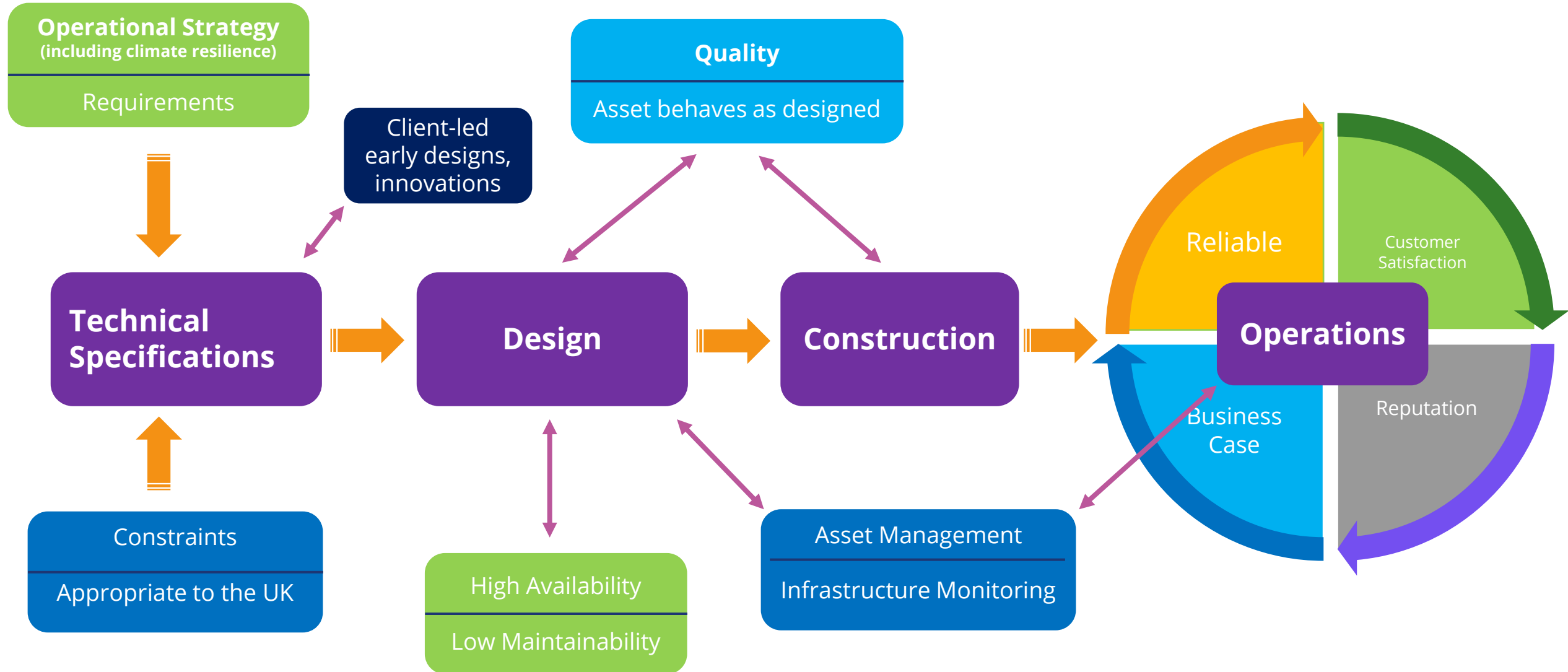
Client-Led Specification and Design

Client organisation should:

- Own the vision, strategy and... the risks
- Collaboration with high speed partners and research institutions
 - Workshops, investigations and reports
 - Objective evidence only
- Incorporate current best practice and lessons from around the world (HS and conventional rail)
- Identify likely failure modes to inform specification and design
- Understand the likely climate change impacts
 - Warmer, wetter winters and hotter, drier summers are predicted in the UK
- Carry out modelling and early design to inform the specification

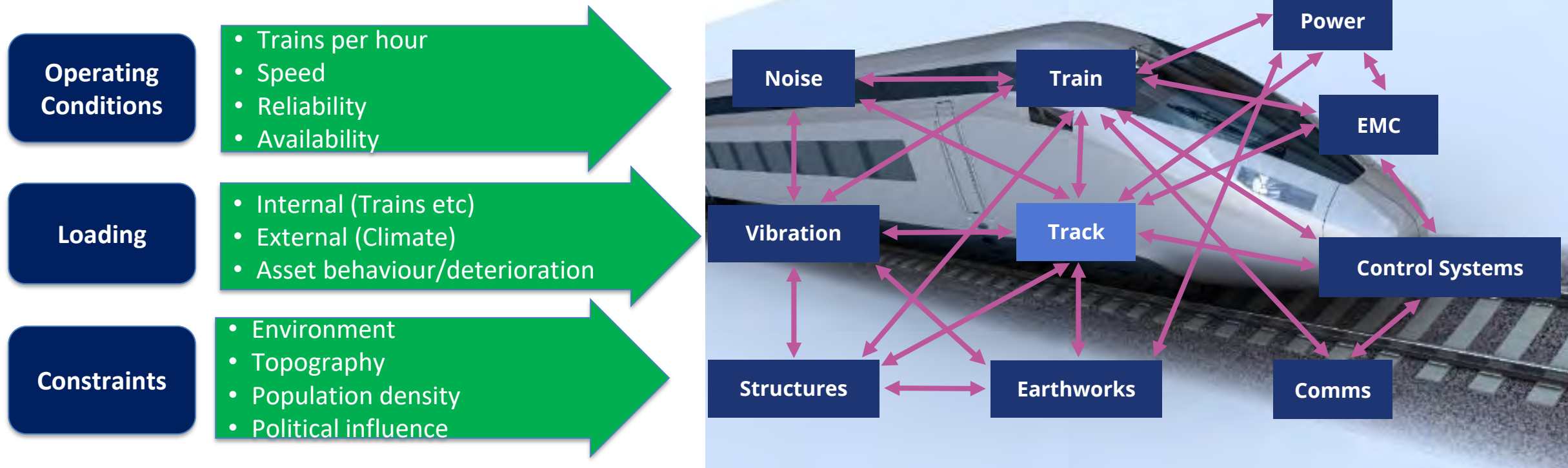


Specifying for a successful outcome...



A Complex Technical System...

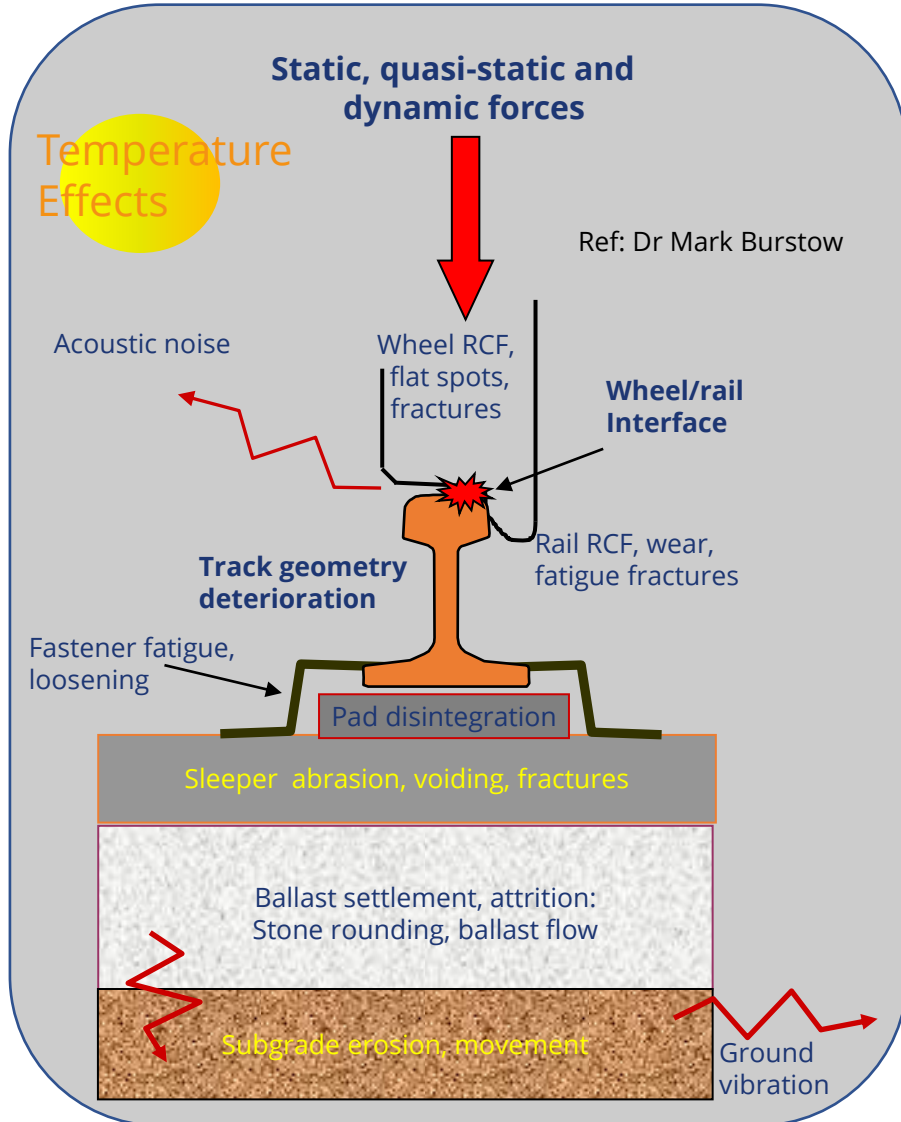
with multiple interfaces



Especially important for a high speed, high capacity railway with stringent reliability targets



Wheel-Rail Interface – key aspect of system design



Forces on the Track System

- All the forces which the track absorbs to support vehicles are transmitted through the wheel-rail interface
- Track damage is a result of energy dissipation from the wheel/rail interface
- The energy is dissipated through various reactions
- Leads to wear, rolling contact fatigue (RCF) and corrugation
- Heavily influenced by track geometry and support stiffness...
...civil engineering structures, especially earthworks

How do we mitigate these effects?

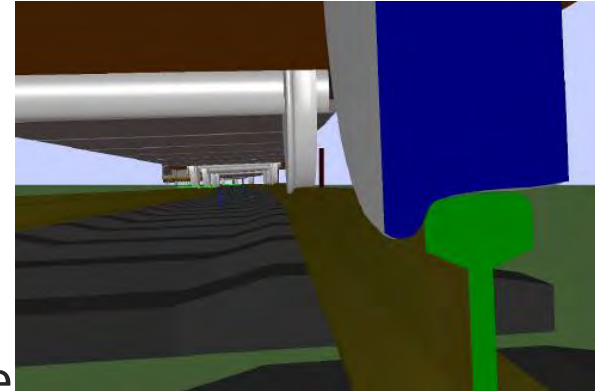
Good specification and design of

Rolling stock – Track system – Support platform



Wheel-Rail Interface – millimetres matter!

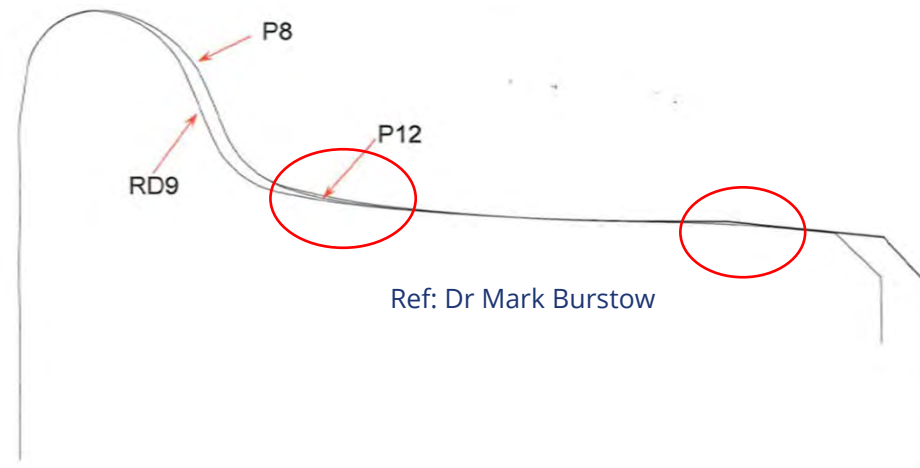
- Wheel-rail contact conditions have a significant affect on the performance of high speed rail systems
- Sub-millimetre tolerances can adverse impact on the wheel-rail interface
 - Wheel profiles, vehicle suspension characteristics
 - Track alignment, rail profiles, localised geometry changes, stiffness changes
- All geometrical parameters for track system and the rolling stock interface must be achieved within specified tolerances



Difference between P8 and P12 wheel profiles < 0.5mm

Significant performance improvement using P12 on WCML

- o Reduction in RCF initiation
- o Better bogie stability
- o Turning intervals increased by 100% (CI390 Pendolinos)



Wheel-Rail Interface – millimetres matter!

- Wheel-rail contact conditions have a significant affect on the performance of high speed rail systems
- Manufacturing tolerances very stringent – rail profiles E1, E2
- Sub-millimetre differences can significantly increase impact forces and contact stresses
- Wheel transfer areas in S&C particularly sensitive
 - Significant increase in vertical impact forces
 - Cast crossing being battered due to 1mm difference in running plane
- Design and manufacture important but also how these relationships are maintained over time

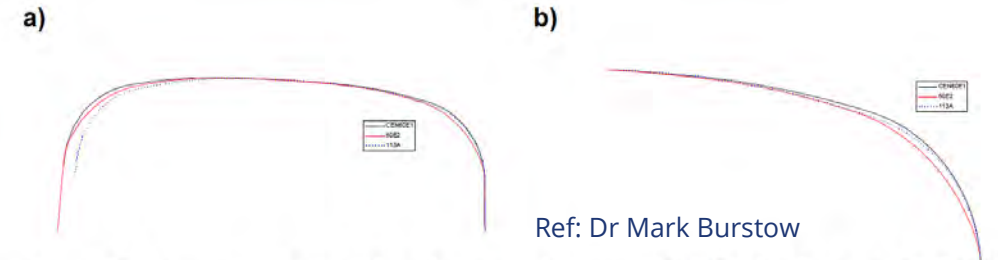


Figure 2: Comparison of rail head shapes between CEN60E1, 60E2 and CEN56E1 (BS113A) for a) width of rail head, and b) the gauge shoulder and gauge corner

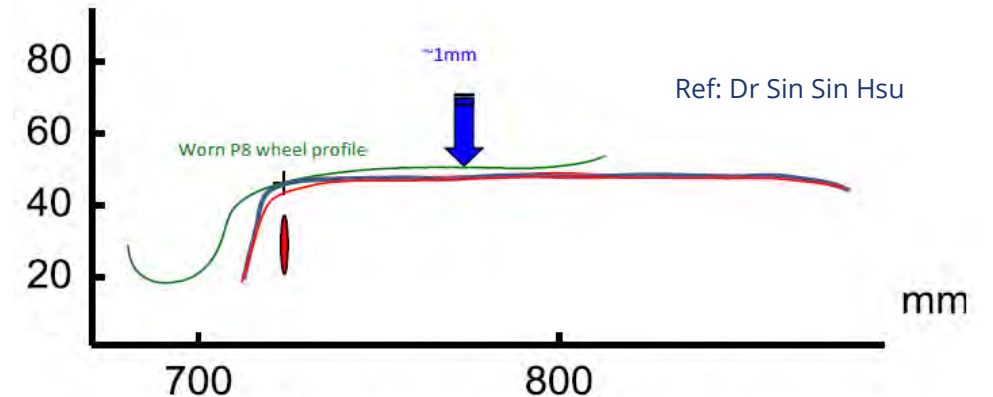
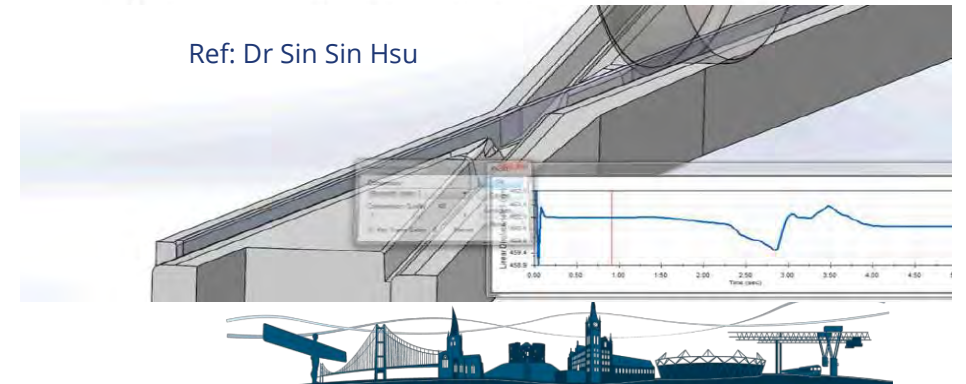


Figure 7: Vertical drop on the running table where the gauge corner

Ref: Dr Sin Sin Hsu

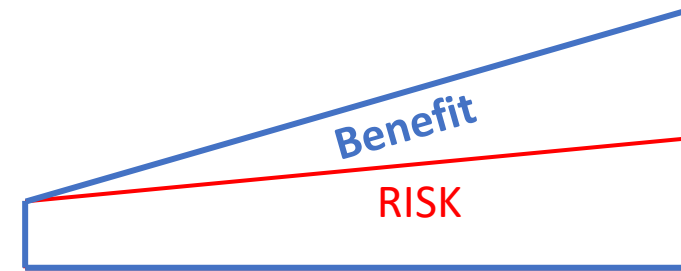


Improving the performance of the infrastructure

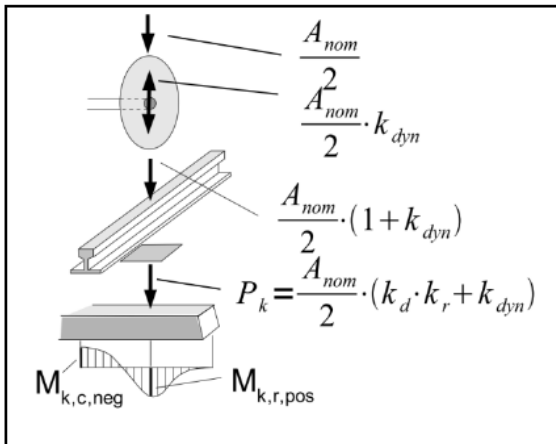
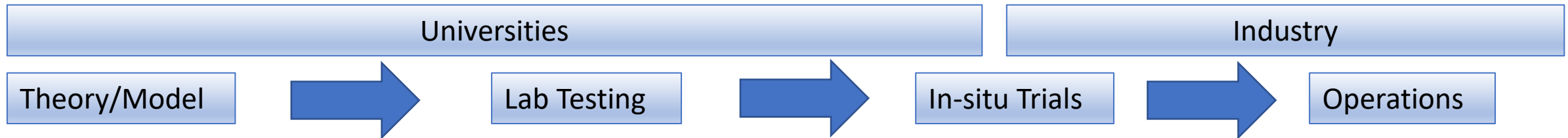
Better reliability and resilience

Demanding operating conditions and high reliability targets at lower costs may require relaxing standards or new technology

Proven
Technology



Research
and
Innovation



Beware the Vagaries of Computing/Modelling

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Can we believe everything we are told?

'Our model tells us we can reduce ballast depths...
reduce transition lengths... save millions with...!'

Various managers/engineers from industry

What is the answer to the 'Ultimate Question of Life,
the Universe and Everything'?

Deep Thought....



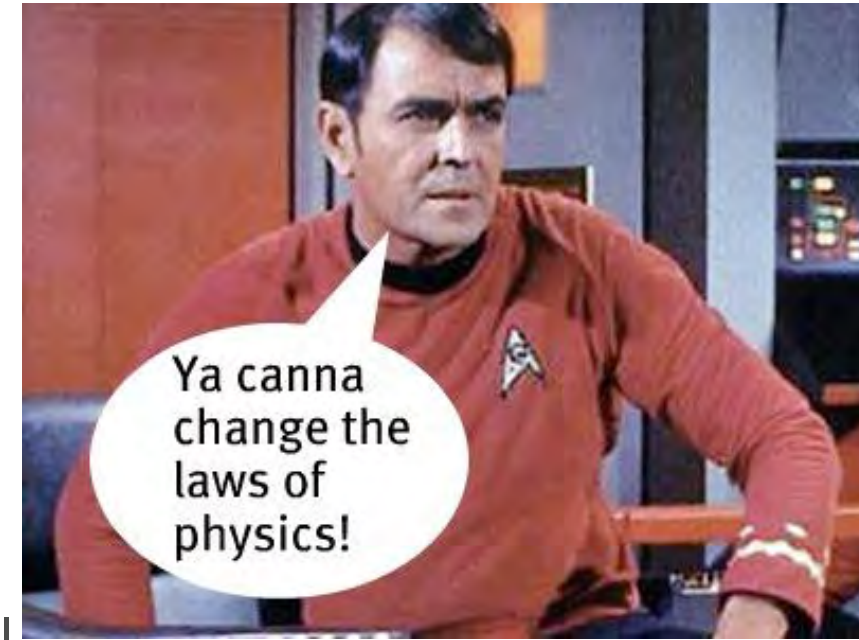
Important that industry has closer and more dynamic relationships with the Academia...

- To understand the capabilities **and** limitations of computers/modelling
- To facilitate access to the operational railway for validation and introduction of new technology



Track Requirements - High Speed (vs conventional?)

- Dedicated passenger only railway (if possible!)
- Good vehicle-track Interaction (classic-compatible trains)
- **Track alignment** suitable for high speed operations
- **Trackform**: Optimised track stiffness and elasticity inc rail deflection
- **Consistent support platform** – earthworks, tunnels, bridges
- Track-bridge interaction design
- Very strict tolerances for geometry, dimensions and wheel-rail profiles, during both construction and maintenance
- Critical-track velocity and geodynamics analysis
- Control of noise and vibrations



E6 – JRE 'classic compatible' train

Application can be different

- System (Components)
- Alignment
- Tolerances, accuracy
- Quality

Focus on integrated system design



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Focus on integrated system design



Star Trek's Scotty for the younger generation



E6 – JRE 'classic compatible' train

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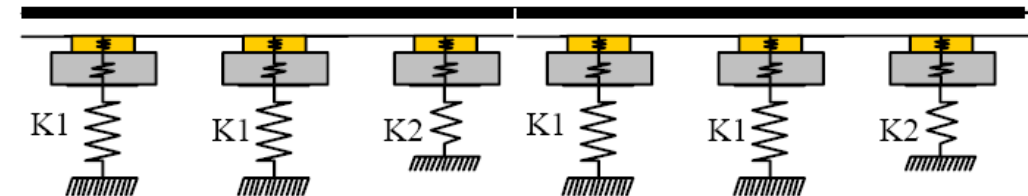
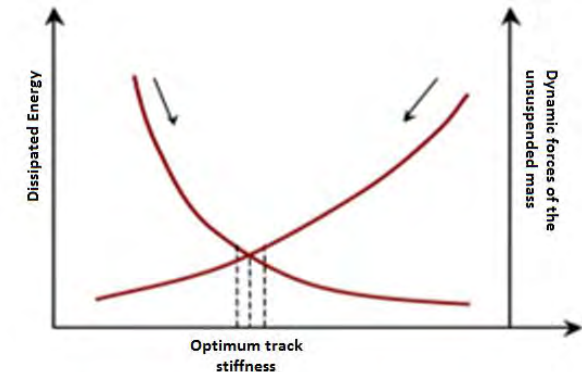
- System (Components)
- Alignment
- Tolerances, accuracy
- Quality



Track System Design for High Speeds - @140mph?

Mechanical behaviour and performance of track enhanced as the speed increases to cope with:

- Higher dynamic loads (axle load x DAF)
 - Higher vibration levels of the rolling stock and superstructure elements (lead to fatigue and wear)
 - Higher stresses on the track bed and support platform/structures
-
- Optimise track stiffness, especially elasticity (vertical and horizontal)
 - Improve load transmission between components
 - Includes the support platform/structures
 - Special trackwork – S&C, Rail Expansion Joints



Case Study: HS2 Choosing the right trackform

Developed a structured evaluation process

Key Inputs

- Technical performance
- Environmental impacts
- Sustainability and resilience
- Operational implications
- Construction programme
- Maintenance & renewals implications
- Capital costs
- Life cycle costs
- Whole life value for money

1. Identify Decisions

2. Define Options & Criteria

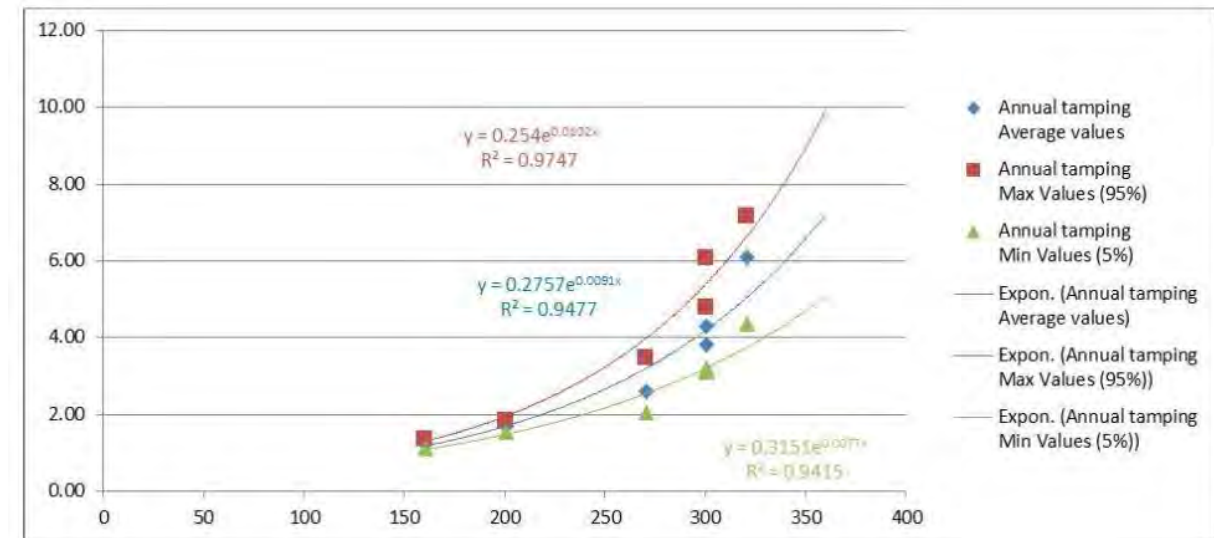
3. Assess Options

4. Evaluate Decision

5a Approval
5b Implement Decision

Statistical analysis by Systra/SNCF of all LGV maintenance databases to predict potential tamping effort for HS2 tonnage and speed

(km)	160 km/h	200 km/h	220 km/h	230 km/h	270 km/h	300 km/h	320 km/h
LN1	6,8 km				283,6 km	484,2	
LN2		14 km	5,6 km		25,6 km	448,8 km	
LN3	0,8 km	14 km	49,8 km	17,8 km	92,2 km	506 km	
LN4						188,6 km	
LN5	0,6 km					317 km	58 km
LN6			0,8 km			84,6 km	475,8 km



Case Study: HS2 Choosing the right trackform

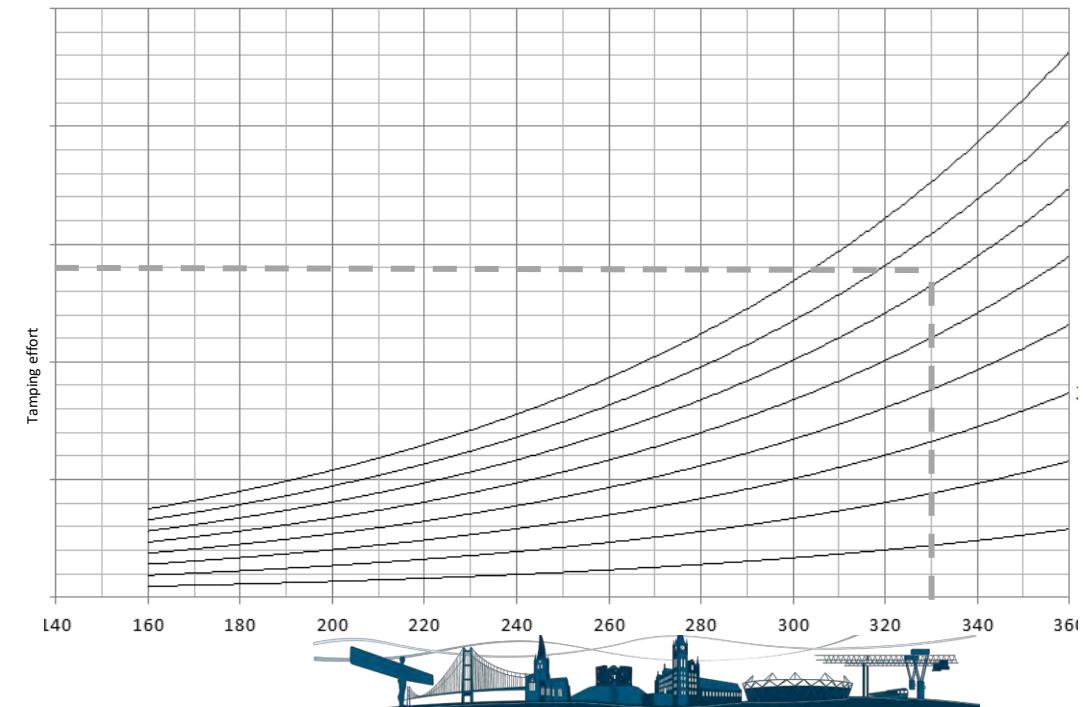
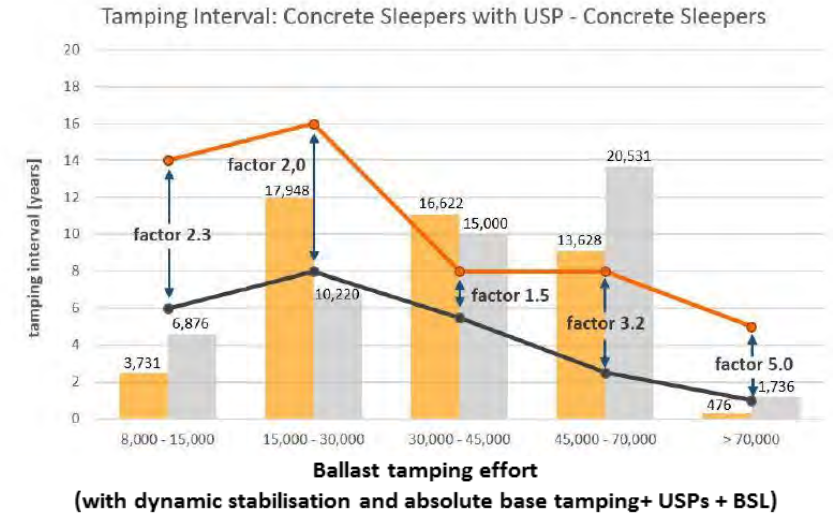
More research to determine performance with bitumen asphalt sub-ballast layer and Under Sleeper Pads

Outputs

- The tamping effort required to maintain ballasted track is a function of tonnage (and speed to a lesser extent)
- Cumulative tonnage is the key input into the degradation of the track system
- The life span of ballast is also a function of the number of tamps...

Higher tonnage = more tamping = lower ballast life = more renewals

Demonstrated that Ballastless Track was better suited to meet the operating strategy and climate resilience

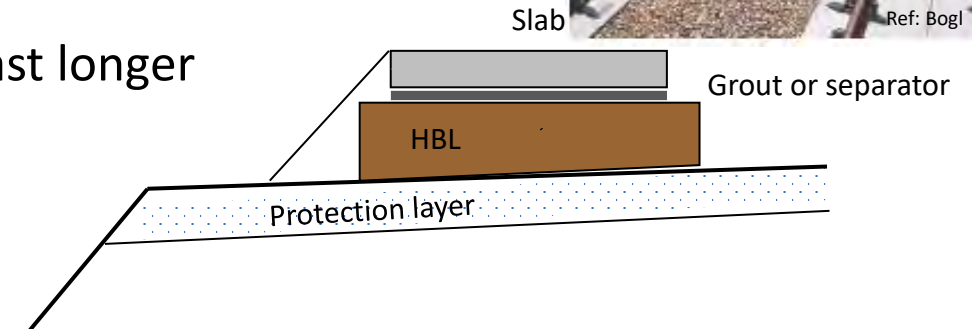


Slab Track : Engineered Design

Pavement
Engineering

Slab track better suited to higher speeds and tonnage due to engineered design

- Engineered design for loading and speed (BS EN 16432)
- Can be modelled and designed using proprietary packages
- Controlled stiffness and rail deflection
- 60+ year design life with minimal maintenance but will last longer
- Fewer risks due to codified design process
- Inherently more stable under loading



Not for spanning poor ground but Earthworks risks can be mitigated

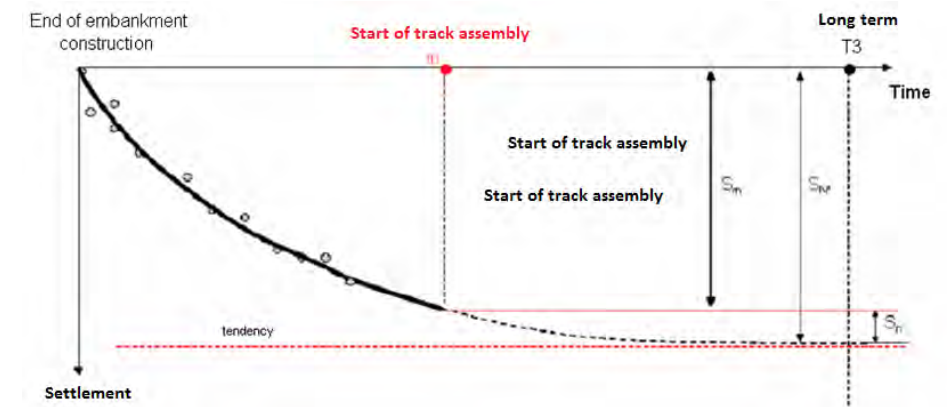
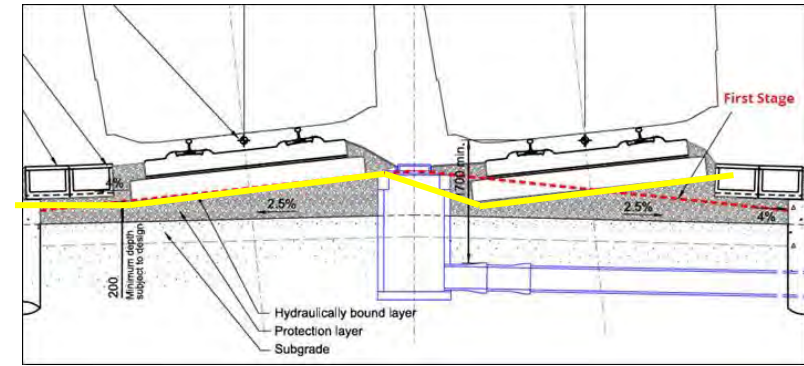
Slab track usually consists of:

- Track slab, rails and fastenings
- Base slab (hydraulic bound layer)
- Subgrade protection layer



Slab Track on Earthworks

- Track slab and HBL/CBL
- Re-profile of the Protection layer where canted
- Track drainage (central drains)
- Predicted ground movements– when to install the track
- Manage on-going settlement and heave



Slab Track : Settlement Mitigation

- Slab track can be corrected when settlement (or heave) occurs
- Remedial options (prioritised in terms of ease)
 1. Adjustable rail fastenings
 2. Jack up the slab units (suits pre-cast)
 3. Grout the earthworks beneath the slab system

Adjustable rail fastenings (approved for HSL)

- Routine method of compensating for ground movement
- Specially designed fastenings and proven in HSLs
- Will accommodate predicted movements (30mm) and unpredicted movements (additional 40mm)
- TSRs not necessarily required, depending on the rate of settlement



Rail height adjustment on Cologne – Frankfurt HSL (operating speed 300km/h)

Adjustable Rail Fastening System

Normal Height regulation

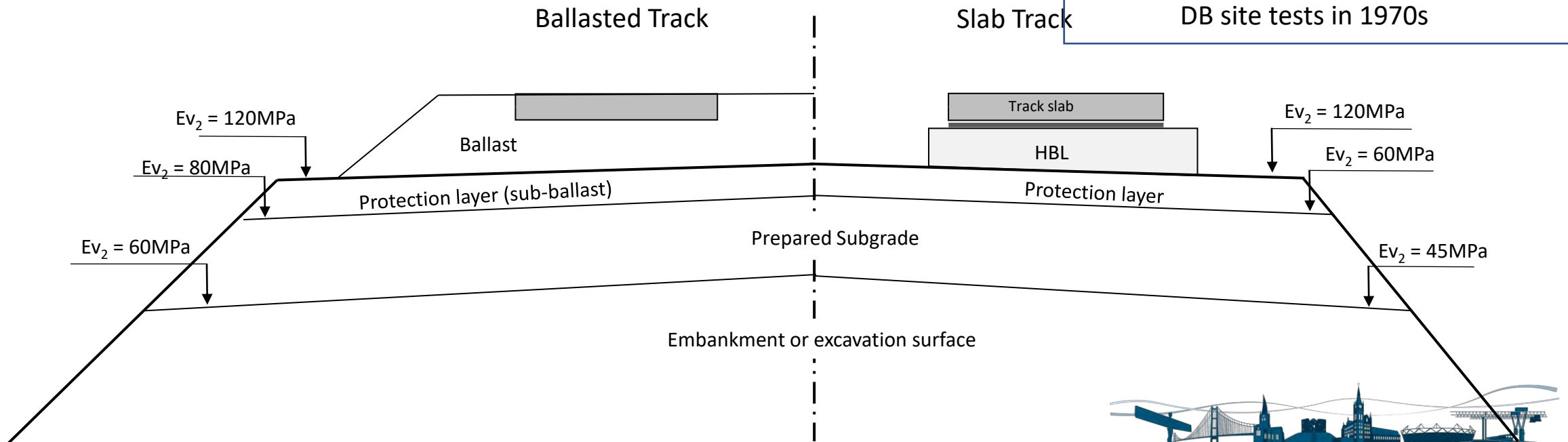
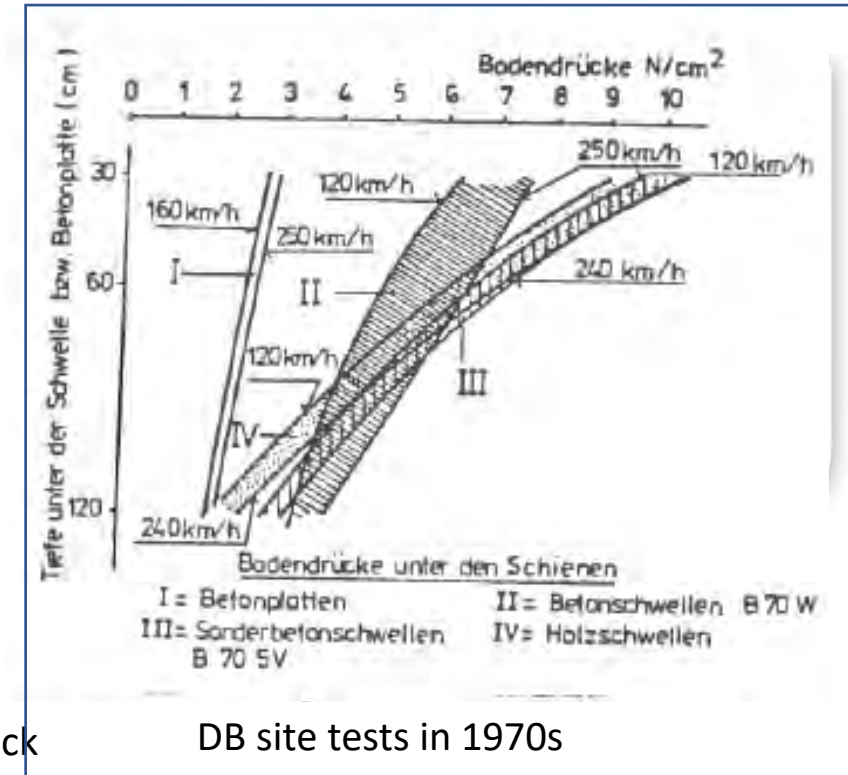
Min: - 4mm

Max: +76



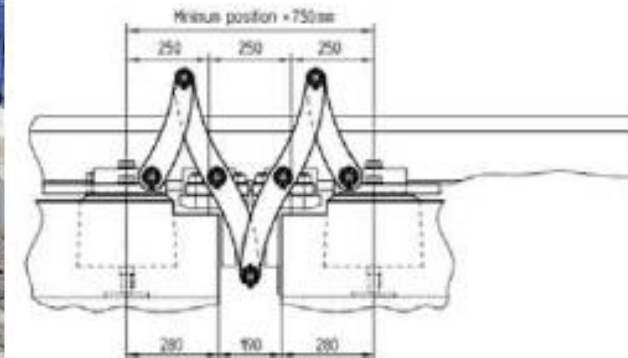
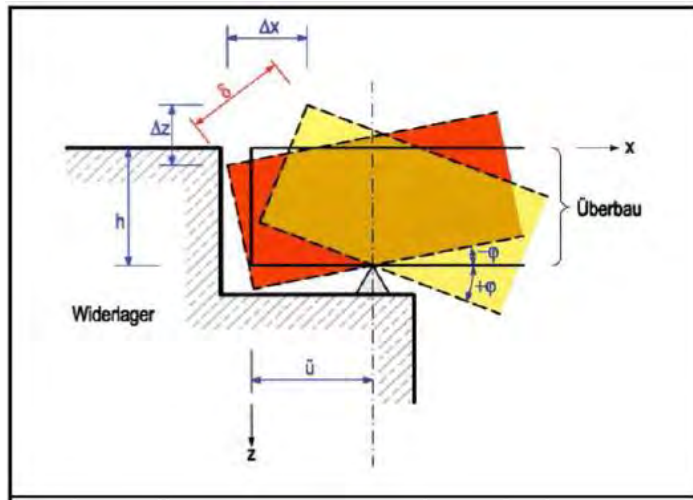
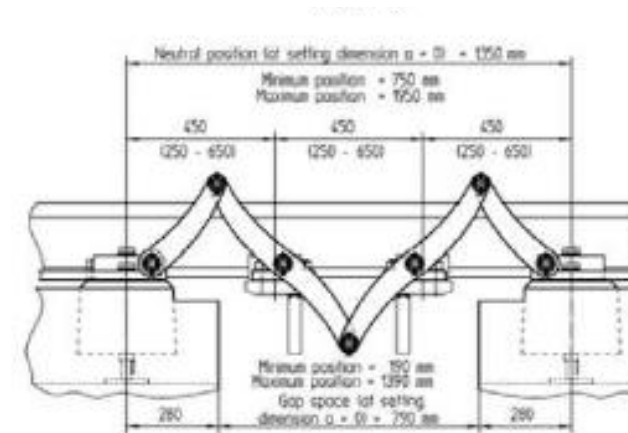
Earthworks for High Speed Rail

- Modern earthworks designs utilise a layered stiffness approach
- Consistent approach worldwide
- Very slight difference between slab and ballast trackforms



Slab Track on Bridges and Viaducts

- Special requirements for slab track on bridges/viaducts
- Cast-in-situ Intermediate Layer
- Track to bridge connection details
- Track-Bridge interaction analysis
- Rail Expansion Devices and comp plates



Slab Track Renewals Implications

- Slab track design life : 60 - 80 years; full renewal : 120 years?
- Individual precast slabs can be replaced (JR West – 5 per night)
- S&C relatively simple as changing the ‘ironwork’ only - individual slabs can be exchanged if required



Full renewal – unknown quantity but unlikely!
Very small percentage of slab units

Safety Implications – slab track

- Implementing a slab track solution will have tangible benefits in terms of workforce health and safety.
- Possibly slight improvement in system safety due to the inherent stability of the slab track system
- Staff safety is significantly improved during construction, maintenance and renewals of the track system
 - Reduced exposure to aggregate dust, HAVS, man-machine interface, slips, trips and falls



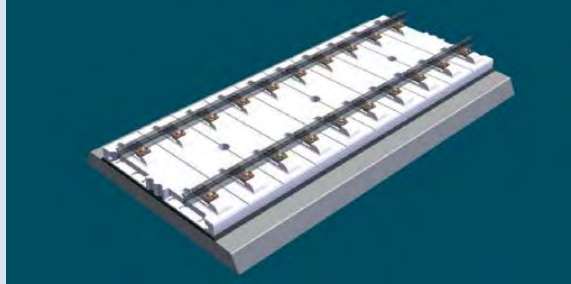
Slab Track Systems

Designs evolved over many years in service

Embedded sleepers in-situ concrete (Rheda/Zublin)



Pre-cast slab track (Bogl)



Pre-cast slab track (OBB Porr)



Resilient track in Tunnels

LVT (Gotthard)

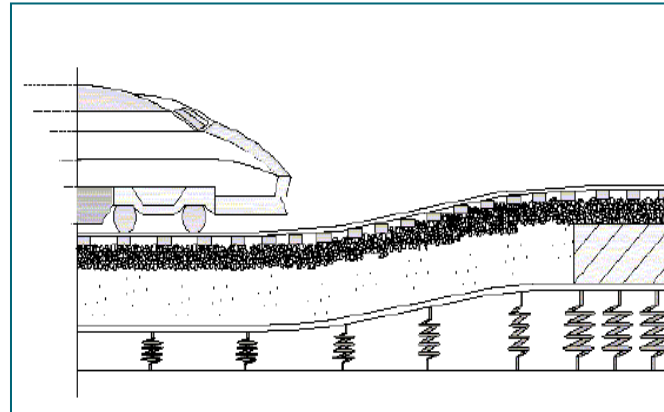
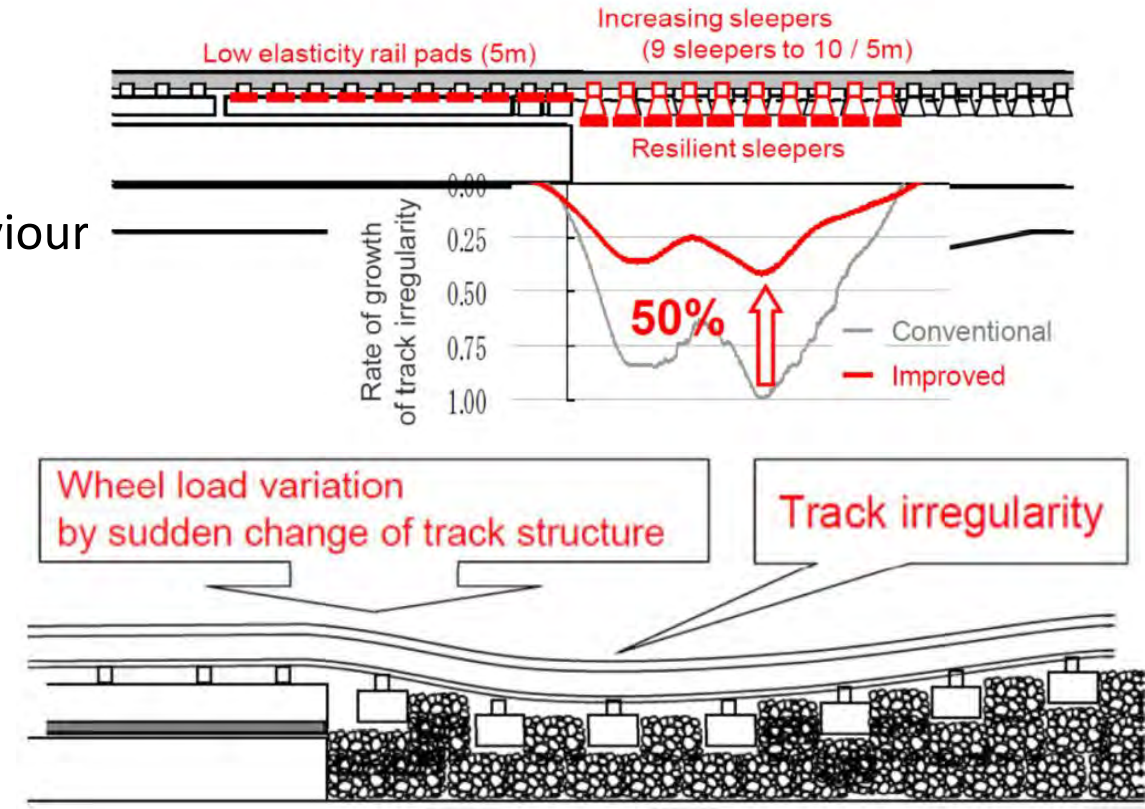


Sateba (HS1)



Slab-ballast transitions

- Maintenance liability that can affect dynamic train behaviour
- Two different issues
 - Variation in stiffness
 - Settlement of the ballast
- Speed limitations (230km/h in France until 2015)
- Optimum designs still evolving
 - Away from dual resilient layers
 - Towards side walls to constrain ballast



Ballasted Track

Typical ballasted
track in Austria

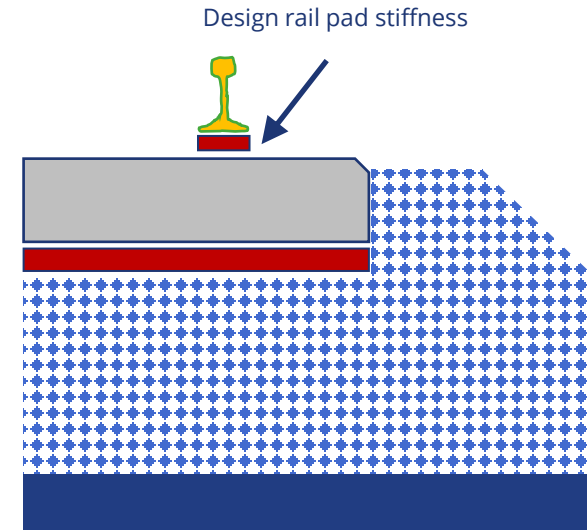
Note the distinct
sub-ballast layer



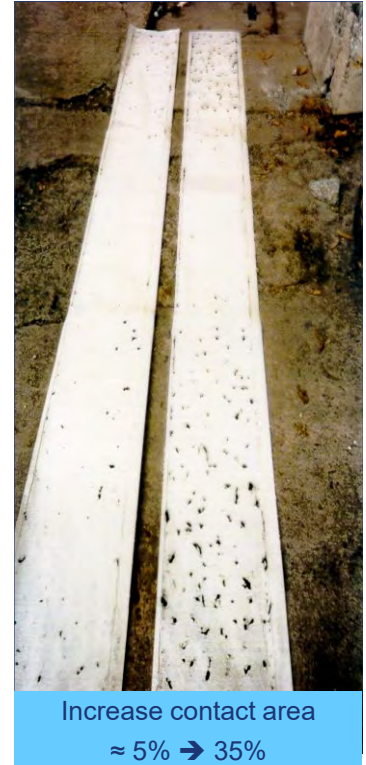
Ballasted Track – Optimisation

OFFICIAL

- Rails – rail head profile, metallurgy
- Fastenings – what design parameters?
- Sleepers
 - Shape, material, behaviour under loading
 - HS sleepers available with 40% lower carbon
 - Dynamic design rather than just axle load
(Rigidity – G44 sleepers not optimised for 140mph speeds)
- Under Sleeper Pads
 - Improve the sleeper/ballast interface and track geometry performance
 - Increase contact area – reduces contact stresses and attrition
 - Reduces initial settlement
 - Designed elasticity and consistent stiffness



$$\delta = F/A$$

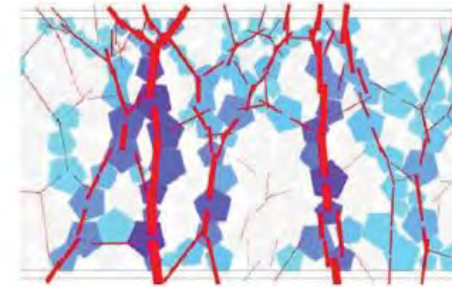
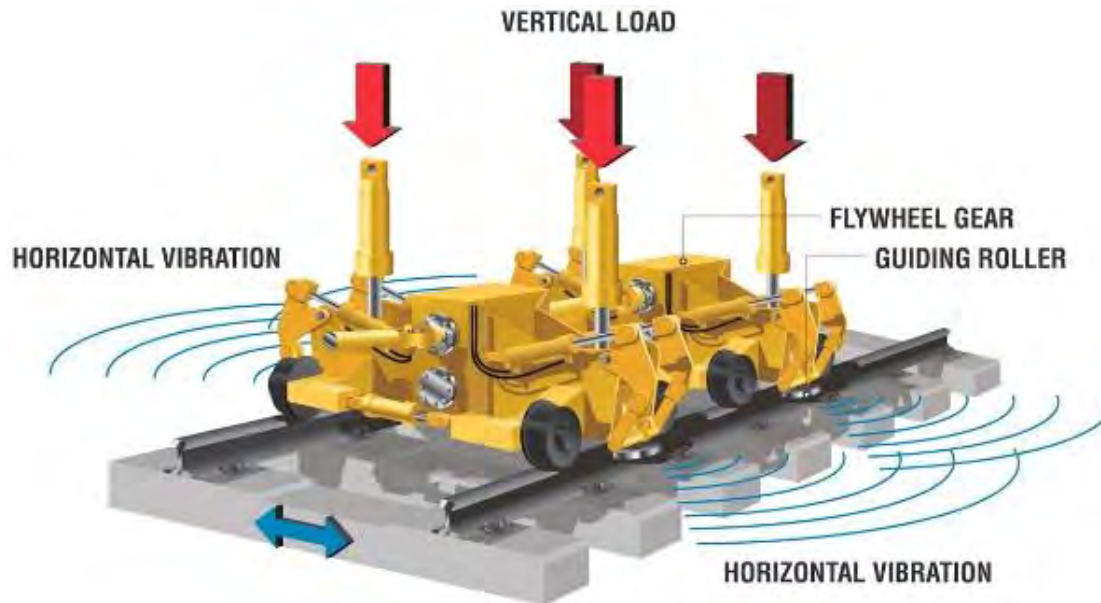


Ballasted Track – Optimisation

Ballast

- High speed spec - LA Co-eff ≤ 14
- Washed, reduced fines
- Consolidated not compacted
- **Installed in layers with DTS...**

...No whacker plates!



Visualisation of "localised rigid structures" that transmit forces in a ballast layer (causine, 2004)

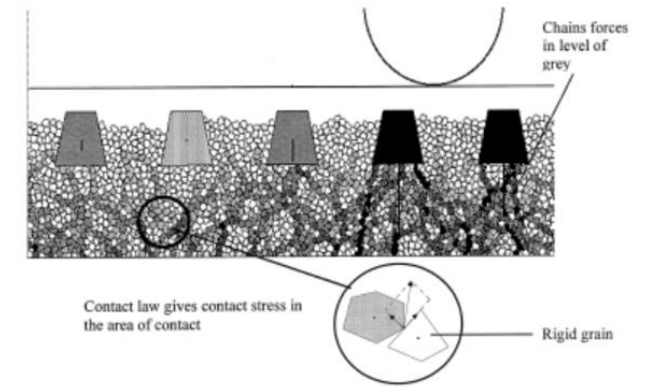


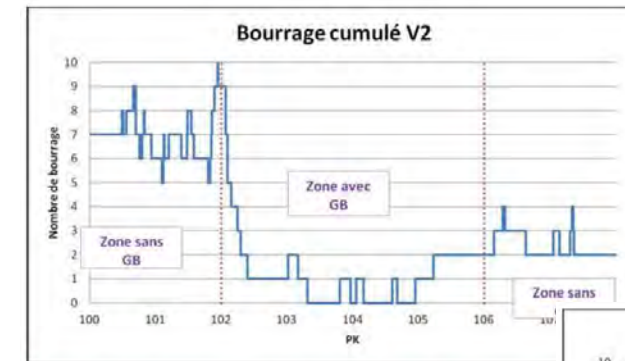
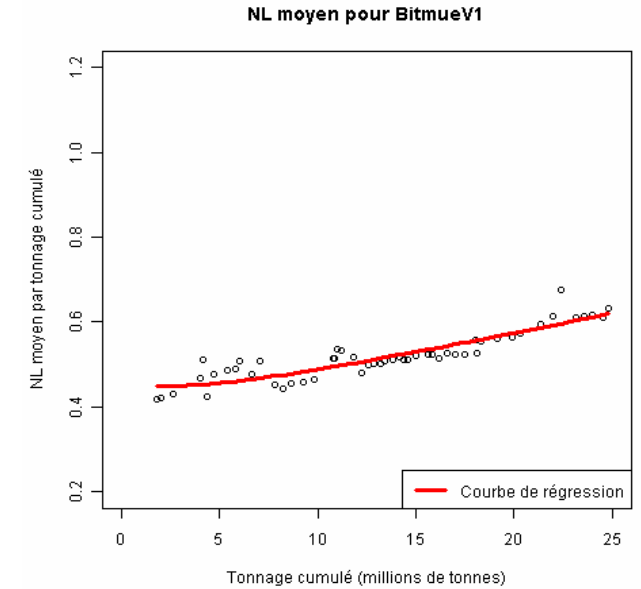
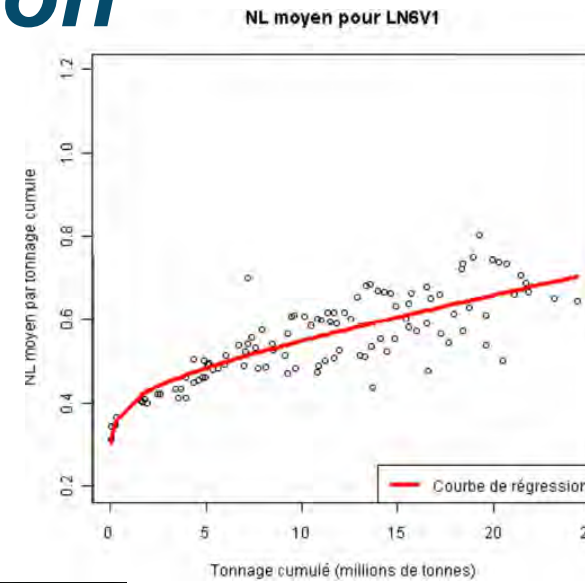
Table 1: Indicative levelling plan.

Work	Levelling and secure (mm)	Definitive difference in RHL (mm)	Cant	Delivery record
Ballast spreading	180	-170		
Previous first levelling	45	-125	1/3 cant definitive	
Stabilized first levelling (100 bars)	-15	-140		Dynamic stabilizer
Second levelling	48	-92		Tamping machine
Second stabilized (75 bars)	-9	-101		Dynamic stabilizer
Third levelling	48	-53	2/3 cant definitive	
Third stabilized (75 bars)	-8	-61		Dynamic stabilizer
Previous reception state (levelling)	40	-21	Definitive cant	Tamping machine
Welding				
Previous reception state (stabilized 100 bars)	-6	-27		Dynamic stabilizer
Neutralisation/ turnout welding				
Reception state (levelling)	29	2		Tamping machine
Reception state (controlled stabilization 30-70 bars)	-2	0		Dynamic stabilizer

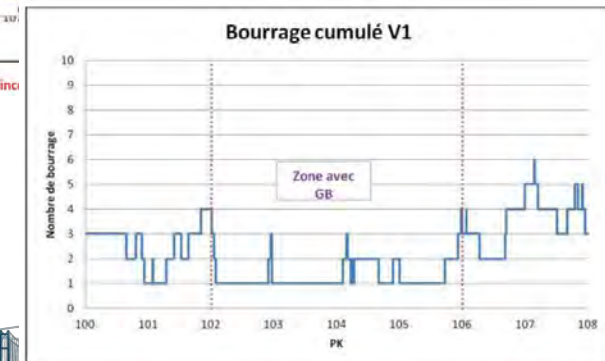
Ballast Track – Optimisation

Sub-ballast Layer

- Bitumen asphalt sub-ballast
 - Uniform stiffness, low deformation (better TQ)
 - Better durability and longer life
 - Waterproof, protects sub-grade
 - Construction advantages



B. Number of tamping operations on a zone with bituminous sublayer (GB in French) since



Number of tamping operations on a zone with bituminous sublayer (GB in French) since 2007 on LN6



Ballast Track – Optimisation

Bitumen asphalt sub-ballast and ballast laid in layers

Nimes – Montpellier HSL

First ballast layer being installed by paving machine on bitumen asphalt sub-ballast

Note groove in the centre of the uncompacted ballast where sleepers and rails will be placed before stabilising the further layers with a DTS



Niall's Ideal Ballasted Track – St Polton to Wien

Design speed = 250km/h

Max operational speed = 230km/h

Ballast retention
(lateral stability and more
durable vertical track
geometry)



Sleepers with USPs
and optimised
shape and length

Heavy duty
concrete
troughing route

Bitumen asphalt sub-ballast
Note: water draining out
under the trough route

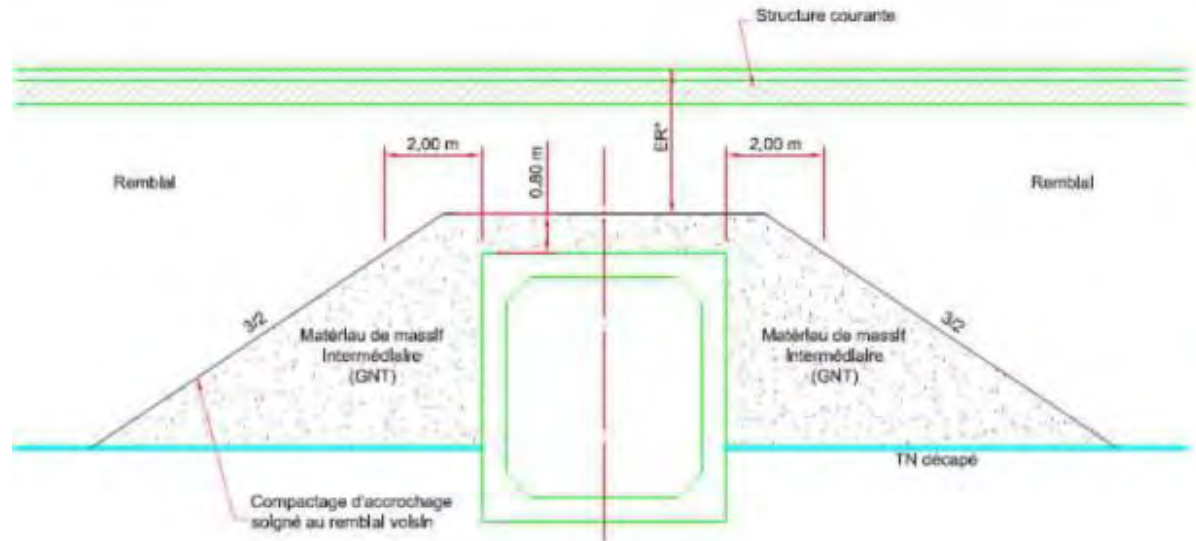


CEN 60 Rails

Ballast laid in layers
with DTS



- Hardspots such as culverts, viaducts, undertrack crossings
- Increased dynamic response of the trains
- Increase in maintenance



High Speed Track Alignment

The design of the track geometry alignment should aim for **good vehicle dynamic behaviour** leading to:

- Better passenger comfort
- Lower dynamic forces → reduced component deterioration
- Improved reliability and availability



Long, flat geometrical elements to limit the number of changes of direction

- Reduce vehicle dynamic response
- Control of lateral and vertical accelerations
- Other operational requirements i.e. power consumption

Consistent application over different areas

Passenger comfort very important on high speed rail

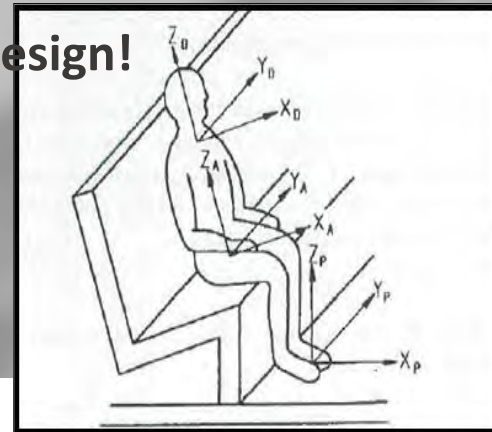


HS2 Track Alignment Design

- Cognisant of and consistent with other HS standards although tailored for HS2 requirements and UK constraints
- Balance between the physical and environmental constraints, costs and a sustainable railway with high passenger satisfaction
- 3 parameter limits: Desirable, Limiting and Exceptional (level of subjectivity in good design)
- Optimum design is not found in the TSI...
- Onus on track alignment designers to consider the impact of their alignment on the train behaviour
- Ultimate objective is to design a safe, reliable and efficient railway

Good design is normally a compliant design...

A simply compliant design is not necessarily a good design!



Some geometry design details – HS2

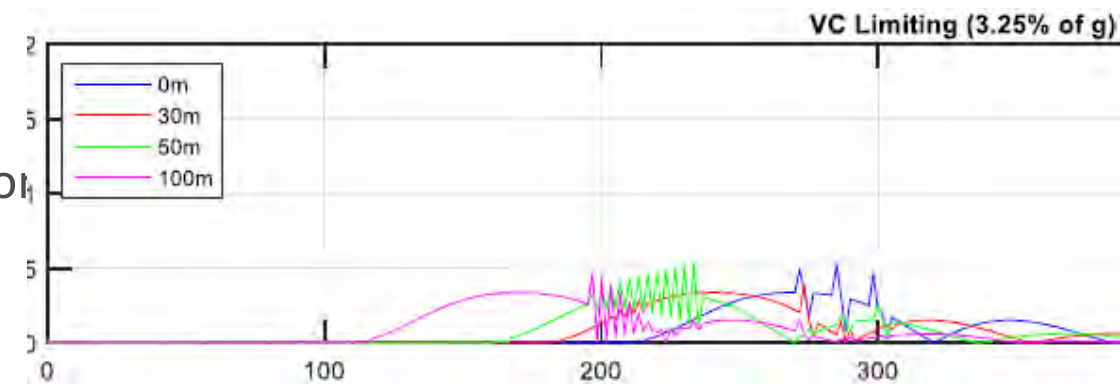
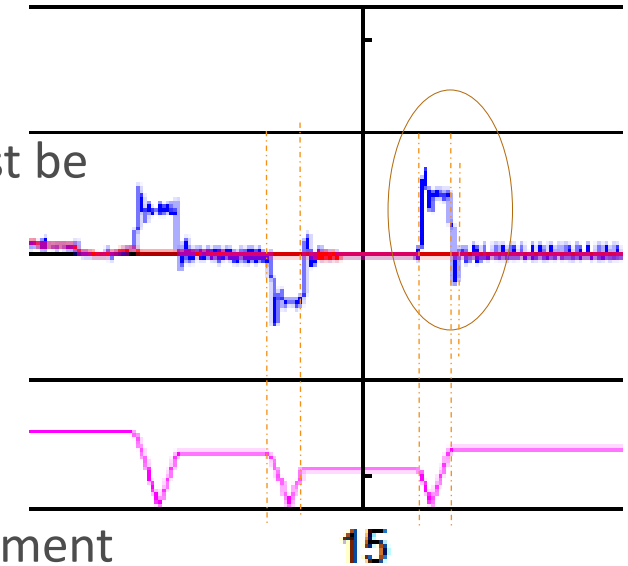
- Alignment optimised for 360km/h but fit for 400km/h where possible (cant/cant deficiency, transitions)
- Typical vertical curve radius 56000m at 400km/h (2.25% g)
 - VC limits: 2.25/3.25/4.25 %g (Desirable/Limiting/Exceptional)
- Horizontal curve radius 8250m or greater (absolute minimum 5910m at 360kph)
- Maximum 4 vertical elements per rolling kilometre to limit changes
- Limiting gradients 2.5%, absolute maximum 3.5%
- Transitions
 - **Clothoid spirals** – these are fine for HSR
 - RoCC/D 30-35 mm/s, 65-70% E+D, 50% at lower speeds
 - Cant gradient 0.25mm/m
- Combined horizontal and vertical curves
 - Normally onerous limits to control accelerations at high speeds
 - Modelled by IRR, Huddersfield, led to relaxations for desirable vertical geometry



Track Alignment Modelling

Combined Horizontal and Vertical Curves

- Both the horizontal and vertical components of high speed track alignment must be considered together due to passenger comfort, buildability and maintainability issues
- Most high speed Infrastructure Managers place some restriction on the use vertical/horizontal transition curves in combination**
- UK topography and population density makes for a challenging high speed alignment design
- Relaxations to certain parameters can help achieve more efficient and cost-effective designs
- IRR, Huddersfield modelled various geometrical combinations (car accelerations and WRI behaviour)
- HS2 analysed outputs and relaxed HS2 requirements for Desirable vertical curves (2.25%g)**

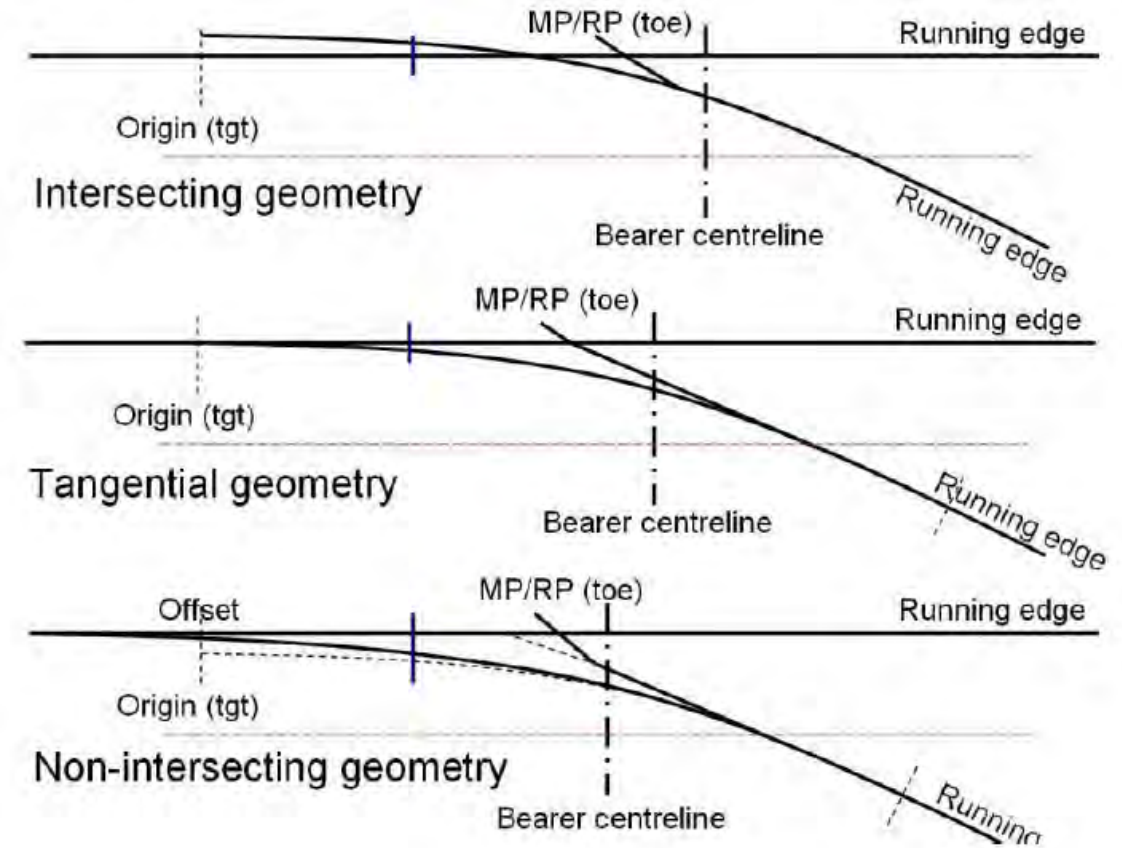


Switches & Crossings



HS2 Switches & Crossings

- Different geometries for high speed and low speed turnouts
- Specially designed for slab track
- Swing nose crossings eliminate impact forces
- Uniform track stiffness through S&C
- Uniform support platform stiffness
- Minimum distance away from bridges



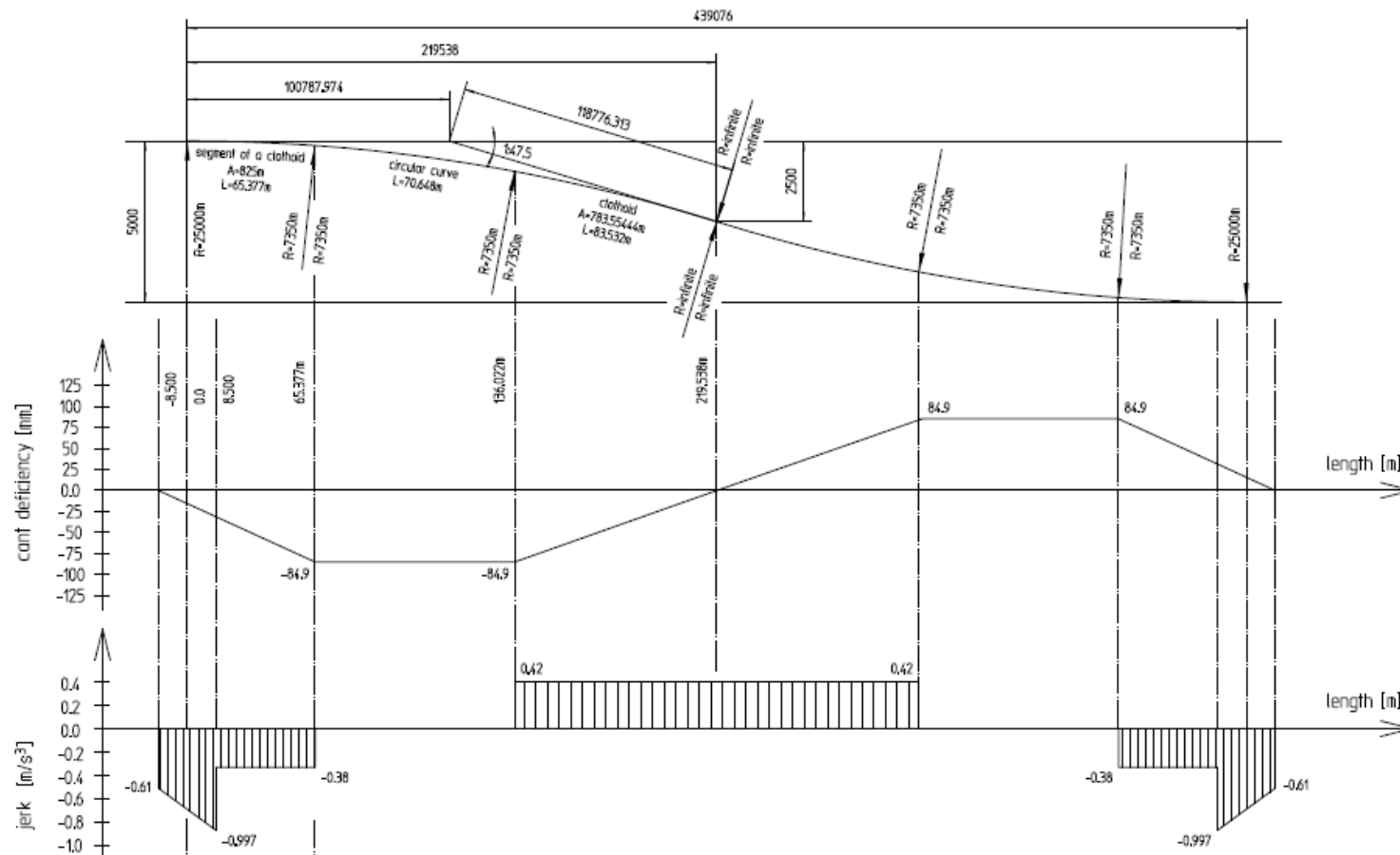
S&C: High performance and low maintenance



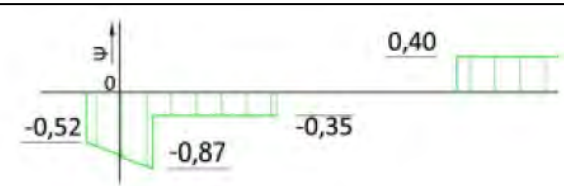
High Speed Switches & Crossings

Double clothoid design

- Geometry optimised for high speeds, low jolt at switch toes
- Sited on straight track
- 450m long crossovers toe-to-toe (230km/h)



Jolt/jerk calculations



Abrupt jerk:

$$\Psi_{\text{abrupt}} = v^3 / ((R_2 - R_1) \times L_{\text{virtual trans}}) \quad (\text{m/s}^3)$$

v : 160km/h (44.44 m/s)

R_1 : 0m (straight)

R_2 : 10000m

$L_{\text{virtual trans}}$: 17m

$$\Rightarrow \Psi_{\text{abrupt}} = 44.44^3 / ((10000 - 0) \times 17) = 0.52 \text{ m/s}^3$$

Constant jerk:

$$\Psi_{\text{constant}} = v^3 / A^2 \quad (\text{m/s}^3)$$

A : 500m

v : 44.44 m/s

$$\Rightarrow \Psi_{\text{constant}} = 44.44^3 / 500^2 = 0.35 \text{ m/s}^3$$

Jerk II:

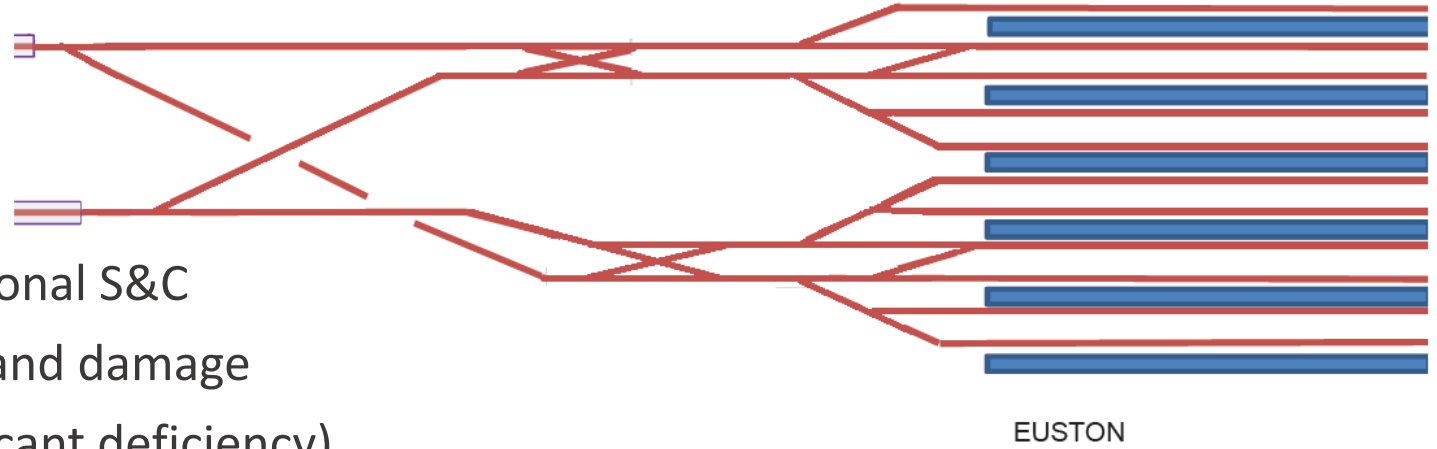
$$\Psi_{\text{II}} = \Psi_{\text{constant}} + \Psi_{\text{abrupt}} \quad (\text{m/s}^3)$$

$$\Rightarrow \Psi = 0.52 + 0.35 = 0.87 \text{ m/s}^3$$



Switches & Crossings (low speed, stations)

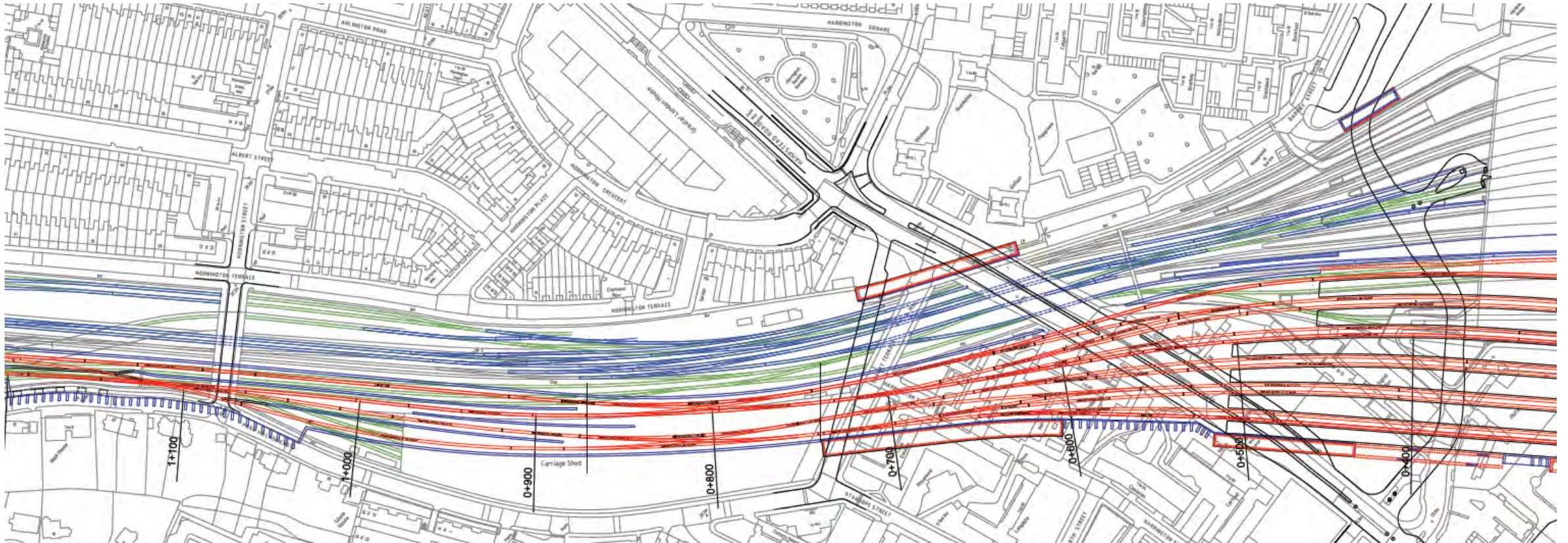
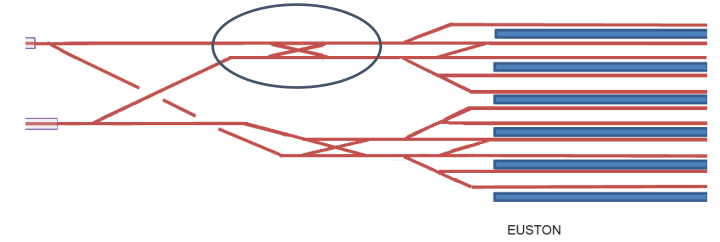
- Constrained footprint in highly urban area
- Congested S&C configuration
- Slab track
- Turnout radius $>400\text{m}$ for all operational S&C
 - Significantly reduces switch wear and damage
 - Curved S&C but no cant (increase cant deficiency)
- No cast crossings – too many failure modes



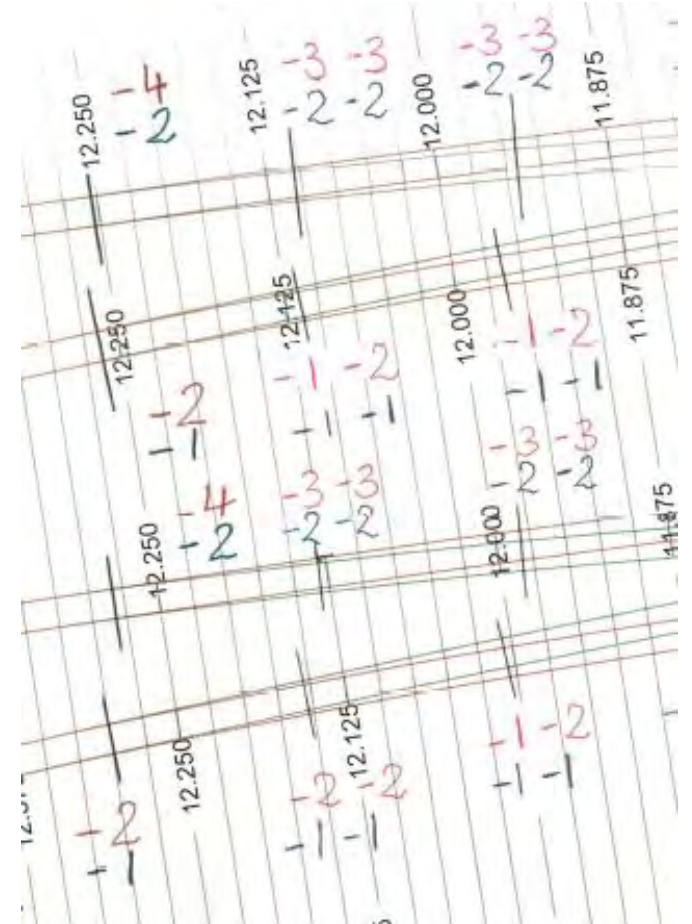
Track Alignment Modelling

Euston Station – Millimetres Matter

- Highly constrained urban area in central London

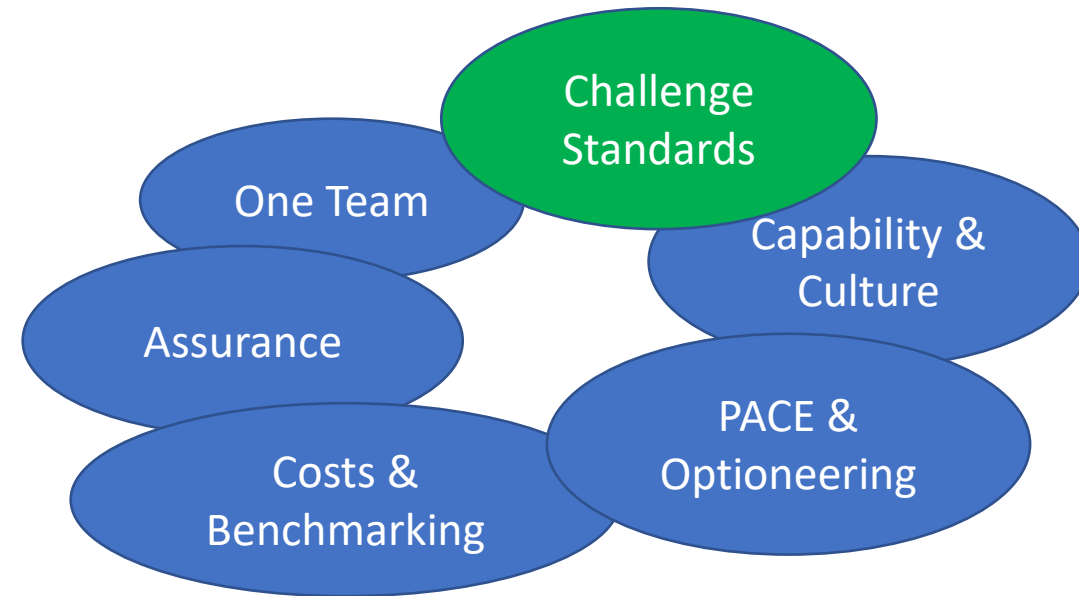


- Height differential between station tracks and approach tunnels (difficult civil engineering works)
- Potential S&C layout on a gradient (1.5%) leading to:
 - cross level errors at toes of scissors curving across gradient
 - Possible impacts on passenger comfort and WRI
- Design method (BRT) amended to remove cross level errors
- Vehicle Dynamics simulations used to confirm no adverse impact to passenger comfort
- Agreed a shallow gradient 0.85%



Current Initiatives in NR – just the beginning

SPEED Initiatives



Minimum Viable Product...!!!

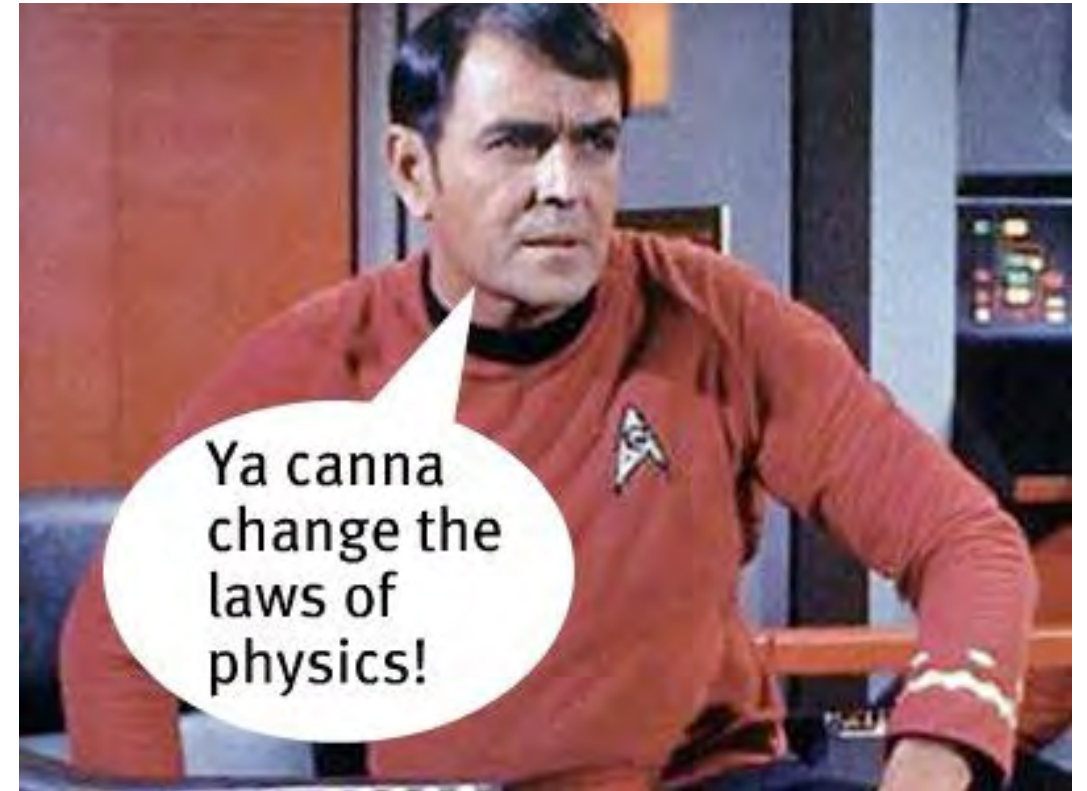
Using trade-offs to achieve an Outcome Based Specification



Engineering Approach to Challenging Standards

- Understand Engineering principles
 - Know the rationale behind the specific clauses/requirements in the standards
 - Understand the risks that are being controlled
 - Evaluate the impact of changing a specific requirement or parameter limit
- (Be numerate when you do so!)

Use the same approach for Innovation



So, what can we do?

As Engineers we have the capability to

- Think critically
- Evaluate
- Decide
- Engineer
- Innovate

Ultimately, engineering is a series of trade-offs

- Required outputs
- Technical knowledge
- Constraints
- Costs vs affordability envelope
- Management of Risk



Thank you for listening



